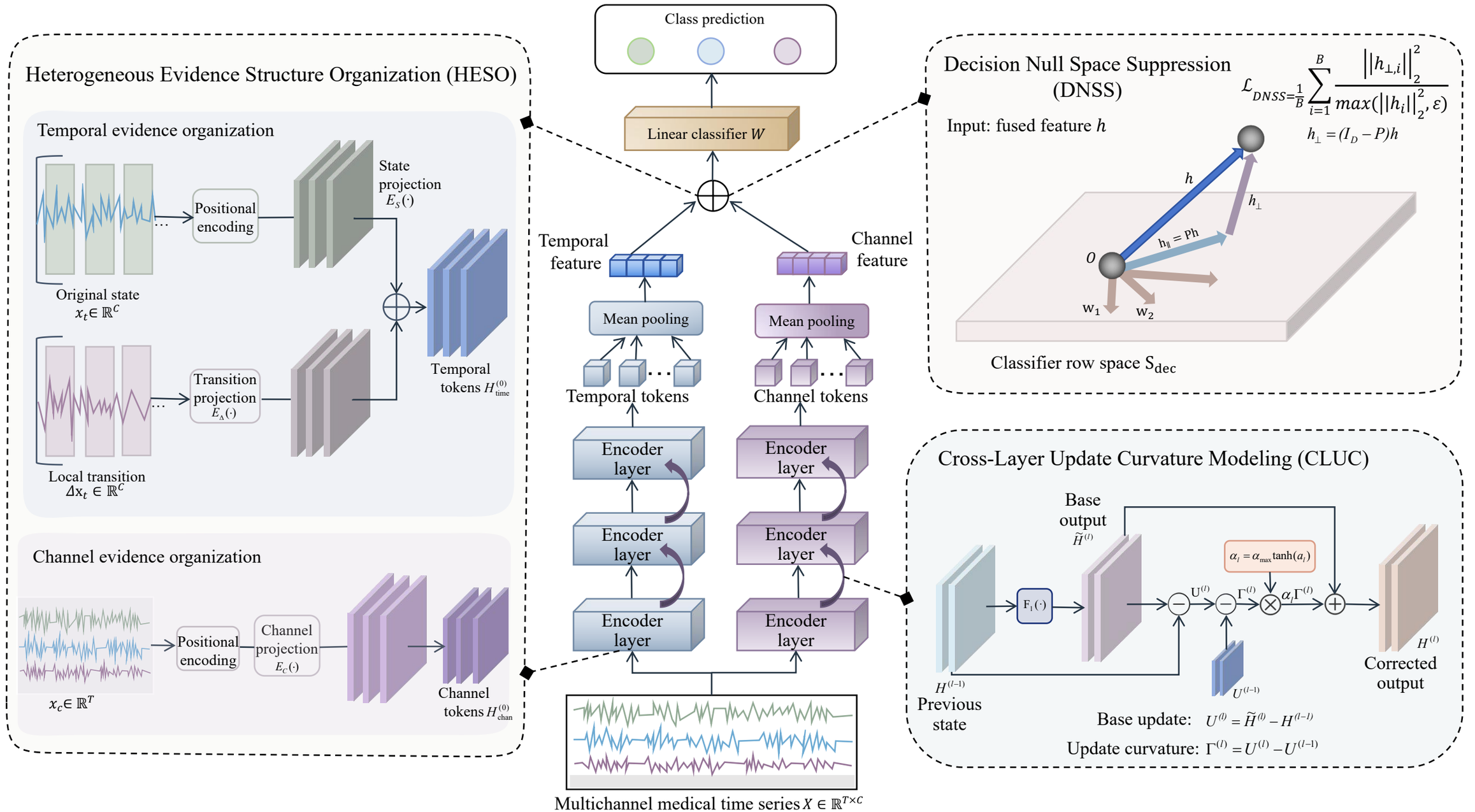

Class prediction
Linear classifier W
Heterogeneous Evidence Structure Organization (HESO)
Temporal evidence organization
Positional encoding
State projection $E_S(\cdot)$
Original state $x_t \in \mathbb{R}^C$
Transition projection $E_\Delta(\cdot)$
Local transition $\Delta x_t \in \mathbb{R}^C$
Temporal tokens $H^{(0)}_{time}$
Channel evidence organization
Positional encoding
Channel projection $E_C(\cdot)$
$x_c \in \mathbb{R}^T$
Channel tokens $H^{(0)}_{chan}$
Temporal feature
Channel feature
Mean pooling
Mean pooling
Temporal tokens
Channel tokens
Encoder layer
Encoder layer
Encoder layer
Encoder layer
Encoder layer
Encoder layer
Multichannel medical time series $X \in \mathbb{R}^{T \times C}$
Decision Null Space Suppression (DNSS)
$\mathcal{L}_{DNSS} = \frac{1}{B}\sum_{i=1}^{B} \frac{\left\|h_{\perp,i}\right\|_2^2}{max(\left\|h_i\right\|_2^2, \varepsilon)}$
Input: fused feature h
$h_\perp = (I_D - P)h$
h
$h_\perp$
$h_\parallel = Ph$
O
w_1
w_2
Classifier row space S_{dec}
Cross-Layer Update Curvature Modeling (CLUC)
Base output $\tilde{H}^{(l)}$
$F_l(\cdot)$
$\alpha_l = \alpha_{max}\tanh(a_l)$
$U^{(l)}$
$\Gamma^{(l)}$
$\alpha_l \Gamma^{(l)}$
$H^{(l)}$
Corrected output
$H^{(l-1)}$
Previous state
$U^{(l-1)}$
Base update: $U^{(l)} = \tilde{H}^{(l)} - H^{(l-1)}$
Update curvature: $\Gamma^{(l)} = U^{(l)} - U^{(l-1)}$